Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
(Deemed to be University Estd. u/s 3 of UGC Act, 1956)

WEEKLY LESSON PLAN

Department: Computer Science and Engineering

Year / Semester: 2025-26 / Winter

Course Code / Course Name: 10213CS224 / Full Stack Application Development

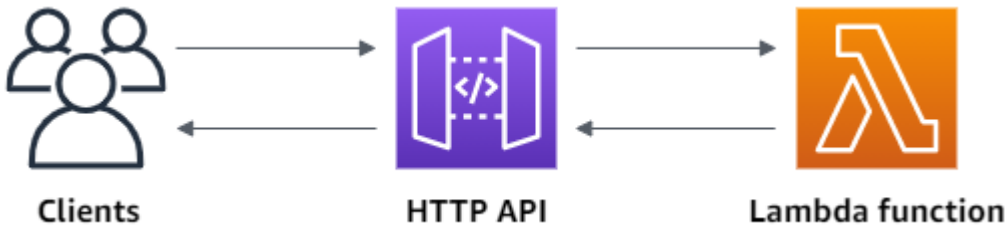
Unit V: Cloud Computing and Vibe Coding **Course Outcome: Build cloud-enabled applications using cloud services and AI-based coding tools.**				
Week	**Session**	**Topics to be Covered**	**Problems to Practice (Hands-On)**	**Resources**
Week 13	Session 61	Cloud Services: API Gateway (Task 22)	**Task** 13.1: Creation of Serverless API Gateway 1. Create a backend of the API using Lambda function. 2. Create an endpoint of Lambda function using HTTP API. 3. Test the API. 4. Clean up the API to prevent unnecessary pricing.	1. https://docs.aws.amazon.com/apigateway/latest/developerguide/getting-started.html 2. https://www.youtube.com/watch?v=C0S01rnN-Os
Week 13	Session 62	Cloud Services: File Storage	**Task** 13.2: Implement file Storage Services for a student named **Arun** is working on a mini project. He wants to store his **project report and images** safely in the cloud so that he can access them from anywhere. Assist him to 1. Create a Storage Space. 2. Uploading Project Files. 3. Access the File Details.	3. https://www.youtube.com/watch?v=vssRTQcewMs 4. https://dev.to/oluwafaj/a-guide-to-creating-a-storage-account-in-microsoft-azure-43ek
Week 13	Session 63	Cloud Services: Compute	**Task** 13.3: Creating and Managing Compute Services 1. Create a Compute Instance (Virtual Machine) 2. Verify Compute Instance Running Status 3. Stop the Compute Instance to Save Cost	1. https://www.youtube.com/watch?v=r6Q_QfmYpLQ 2. https://dev.to/eunice-js/aws-compute-services-the-complete-guide-to-ec2-and-beyond-4mjc

				3. https://www.youtube.com/watch?v=kogSnxk-5t0
Week 13	Session 64	Cloud Services: Relational Database Services	**Task** 13.4: Relational Database Service (RDS) 1. Create a relational database service suitable for the HealthTrack application. 2. Verify that the RDS instance is active and ready to store data. 3. Check Backup and Maintenance Settings	1. https://www.youtube.com/watch?v=-p0P12HboA0 2. https://dev.to/s3cloudhub/aws-relational-database-how-to-create-rds-instance-2dn6
Week 13	Session 65	Cloud Services: Pay-As-You-Use	**Task** 13.5: Utilize Pay-As-You-Use Cloud Model for a startup named **EduFlex** is launching an online learning platform. Initially, they expect **low traffic**, but during exam seasons the number of users may suddenly increase. The startup has limited capital and wants to avoid heavy upfront infrastructure costs. 1. Explain why **Pay-As-You-Use** cloud model is suitable for EduFlex. 2. Identify cloud services EduFlex should use instead of on-premise servers. 3. Estimate cost benefits when: ○ Normal days → 200 users ○ Peak days → 10,000 users	https://www.youtube.com/watch?v=QA5srDUJCjY

Solution:

Task 13.1: Creation of Serverless API Gateway

1. Create a backend of the API using Lambda function.
2. Create an endpoint of Lambda function using HTTP API.
3. Test the API.
4. Clean up the API to prevent unnecessary pricing.



To complete this exercise, you need an AWS account and an AWS Identity and Access Management user with console access. For more information, see Set up to use API Gateway.

You use a Lambda function for the backend of your API. Lambda runs your code only when needed and scales automatically, from a few requests per day to thousands per second.
For this example, you use the default Node.js function from the Lambda console.

To create a Lambda function

1. Sign in to the Lambda console at https://console.aws.amazon.com/lambda.
2. Choose **Create function**.
3. For **Function name**, enter **my-function**.
4. For all other options, use the default setting.
5. Choose **Create function**.

The example function returns a 200 response to clients, and the text Hello from Lambda!.
You can modify your Lambda function, as long as the function's response aligns with the format that API Gateway requires.
The default Lambda function code should look similar to the following:

```
export const handler = async (event) => {
  const response = {
    statusCode: 200,
    body: JSON.stringify('Hello from Lambda!'),
  };
  return response;
};
```

Step 2: Create an HTTP API

Next, you create an HTTP API. API Gateway also supports REST APIs and WebSocket APIs, but an HTTP API is the best choice for this exercise. REST APIs support more features than HTTP APIs, but we don't need those features for this exercise. HTTP APIs are designed with minimal features so that they can be offered at a lower price. WebSocket APIs maintain persistent connections with clients for full-duplex communication, which isn't required for this example.

The HTTP API provides an HTTP endpoint for your Lambda function. API Gateway routes requests to your Lambda function, and then returns the function's response to clients.

To create an HTTP API

1. Sign in to the API Gateway console at https://console.aws.amazon.com/apigateway.
2. Do one of the following:
    - To create your first API, for **HTTP API**, choose **Build**.
    - If you've created an API before, choose **Create API**, and then choose **Build** for **HTTP API**.
3. For **Integrations**, choose **Add integration**.
4. Choose **Lambda**.
5. For **Lambda function**, enter **my-function**.
6. For **API name**, enter **my-http-api**.
7. Choose **Next**.
8. Review the *route* that API Gateway creates for you, and then choose **Next**.
9. Review the *stage* that API Gateway creates for you, and then choose **Next**.
10. Choose **Create**.

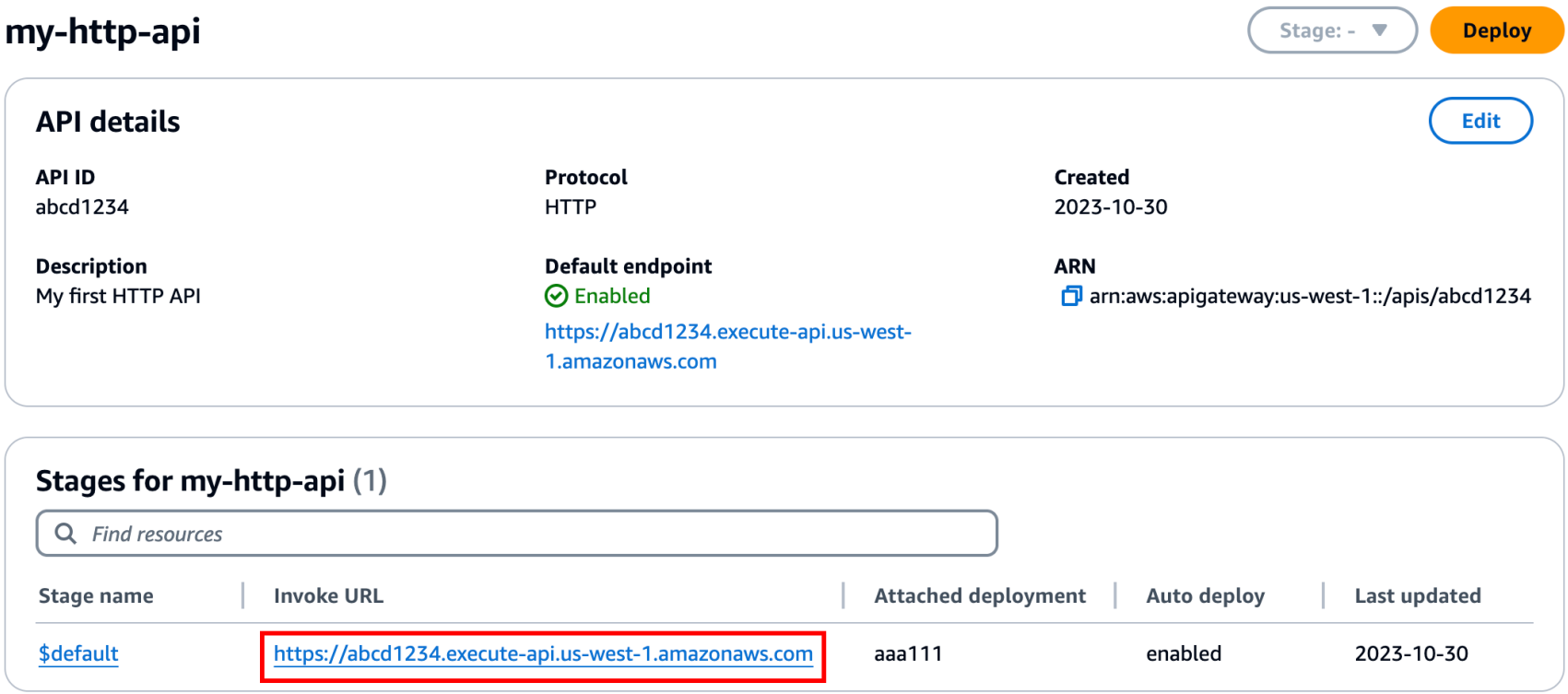
Now you've created an HTTP API with a Lambda integration that's ready to receive requests from clients.

Step 3: Test your API

Next, you test your API to make sure that it's working. For simplicity, use a web browser to invoke your API.

To test your API

1. Sign in to the API Gateway console at https://console.aws.amazon.com/apigateway.
2. Choose your API.
3. Note your API's invoke URL.



4. Copy your API's invoke URL, and enter it in a web browser. Append the name of your Lambda function to your invoke URL to call your Lambda function. By default, the API Gateway console creates a route with the same name as your Lambda function, my-function.

The full URL should look like https://*abcdef123*.execute-api.*us-east-2*.amazonaws.com/*my-function*.

Your browser sends a GET request to the API.

5. Verify your API's response. You should see the text "Hello from Lambda!" in your browser.

(Optional) Step 4: Clean up

To prevent unnecessary costs, delete the resources that you created as part of this getting started exercise. The following steps delete your HTTP API, your Lambda function, and associated resources.

To delete an HTTP API

1. Sign in to the API Gateway console at https://console.aws.amazon.com/apigateway.

2. On the **APIs** page, select an API. Choose **Actions**, and then choose **Delete**.
3. Choose **Delete**.

To delete a Lambda function

1. Sign in to the Lambda console at https://console.aws.amazon.com/lambda.
2. On the **Functions** page, select a function. Choose **Actions**, and then choose **Delete**.
3. Choose **Delete**.

To delete a Lambda function's log group

1. In the Amazon CloudWatch console, open the Log groups page.
2. On the **Log groups** page, select the function's log group (/aws/lambda/my-function). Choose **Actions**, and then choose **Delete log group**.
3. Choose **Delete**.

To delete a Lambda function's execution role

1. In the AWS Identity and Access Management console, open the Roles page.
2. Select the function's role, for example, my-function-*31exxmpl*.
3. Choose **Delete role**.
4. Choose **Yes, delete**.

.

Task 13.2: File Storage Services

Scenario

A student named **Arun** is working on a mini project. He wants to store his **project report and images** safely in the cloud so that he can access them from anywhere.

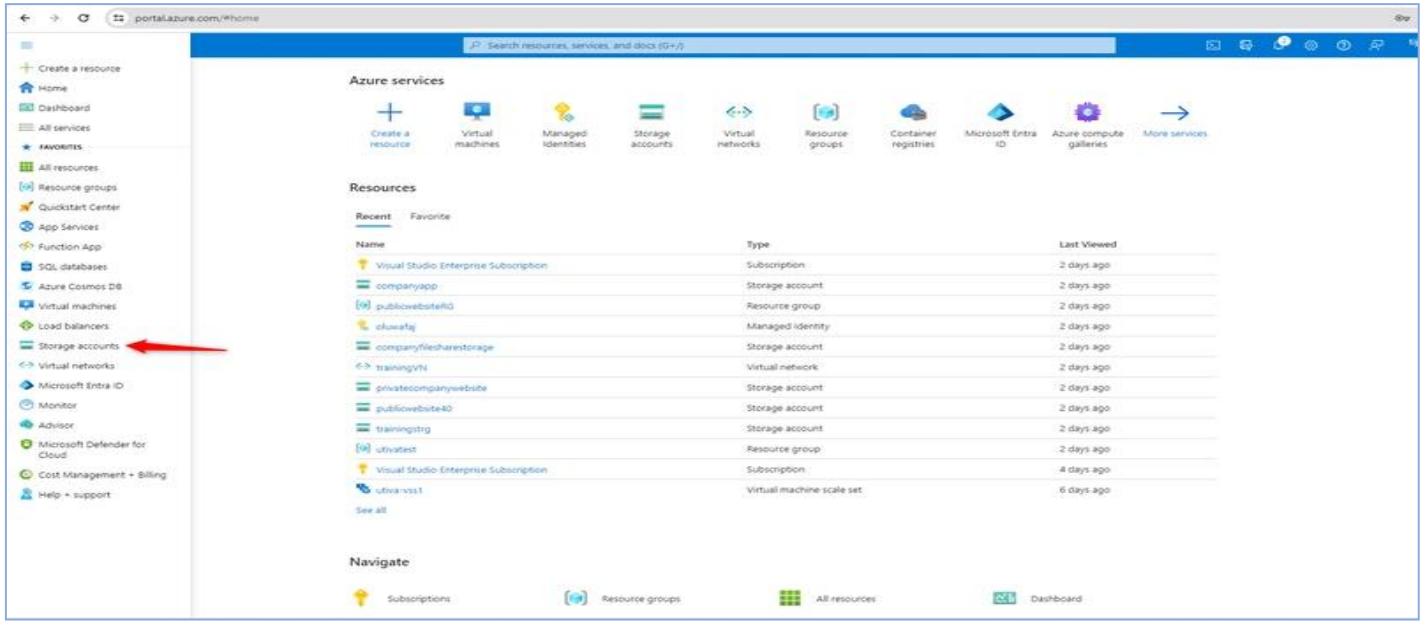
1. Create a Storage Space.
2. Uploading Project Files.
3. Access the File Details.

Step 1: Sign in to Azure Portal:

- Begin by navigating to the Azure Portal and sign in with your Azure account credentials.

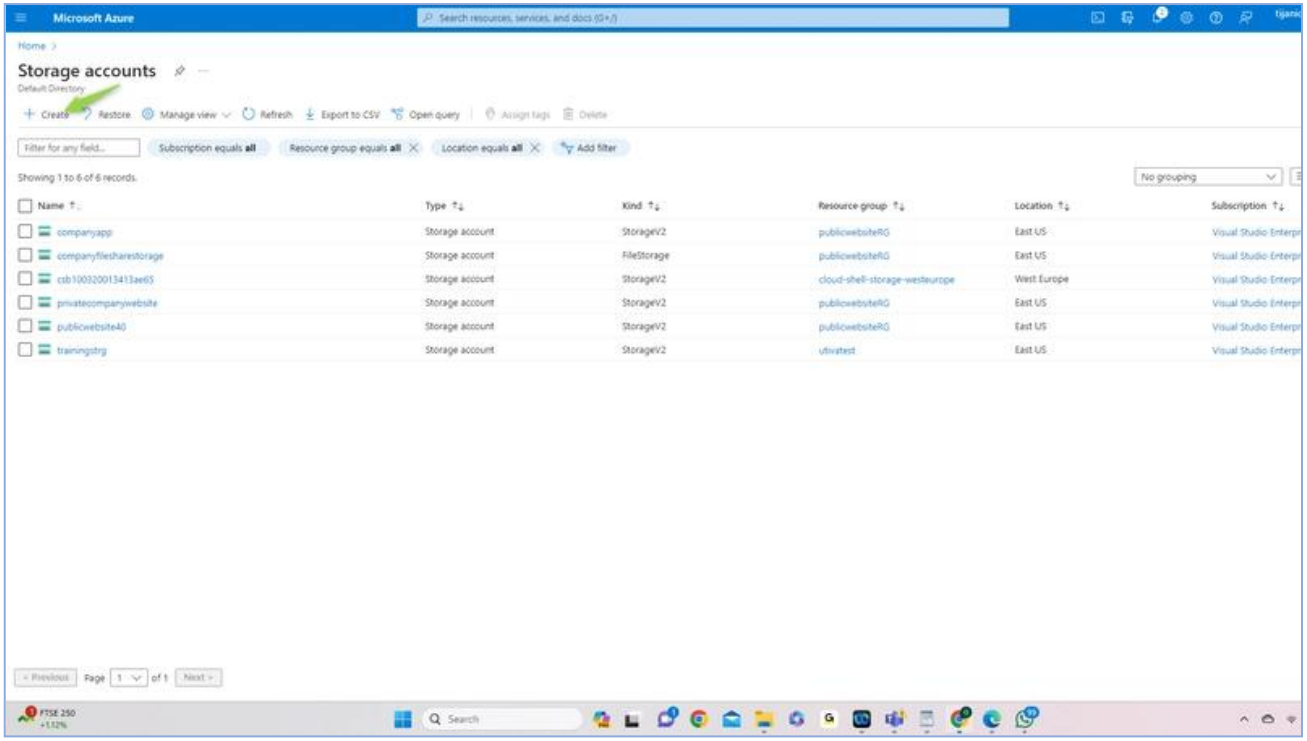
Step 2: Access the Storage Accounts Section:

- In the left-hand menu, click on "Storage accounts" to access the storage management section.



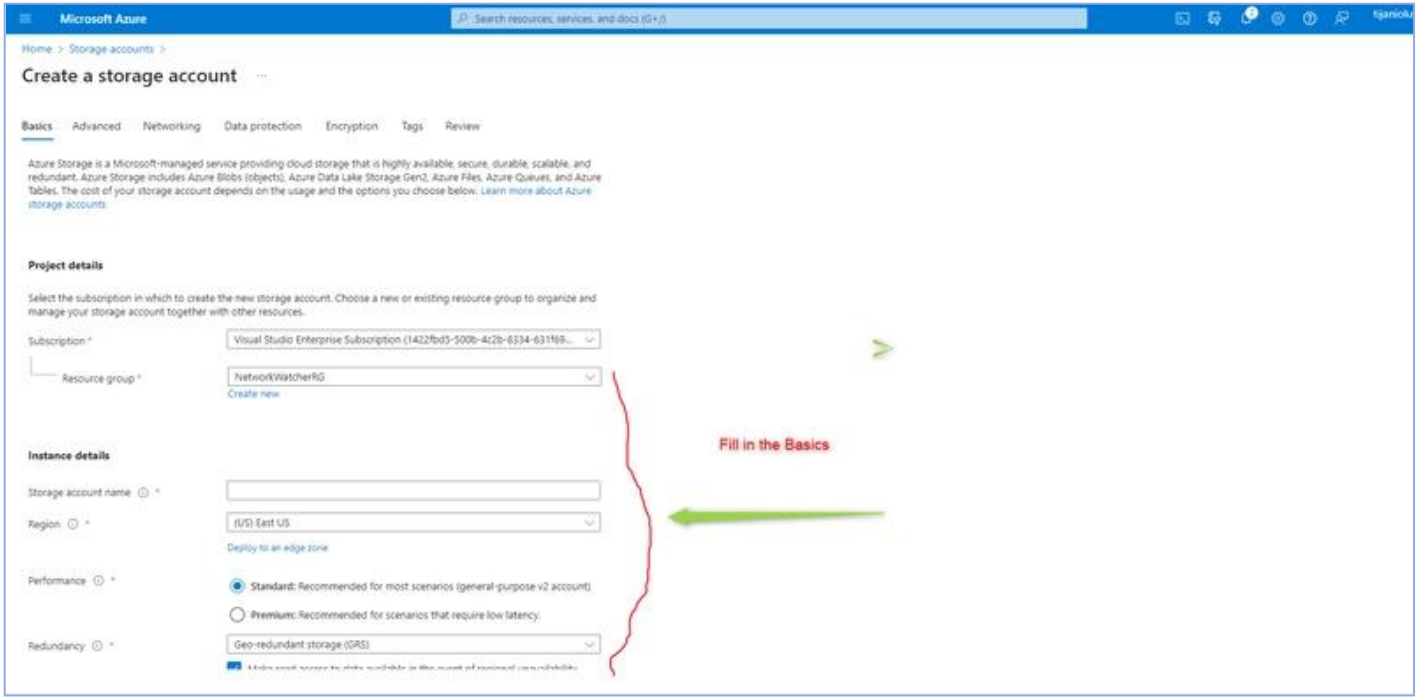
Step 3: Create a New Storage Account:

- Click on the "+ Add" button to start the process of creating a new Storage Account.



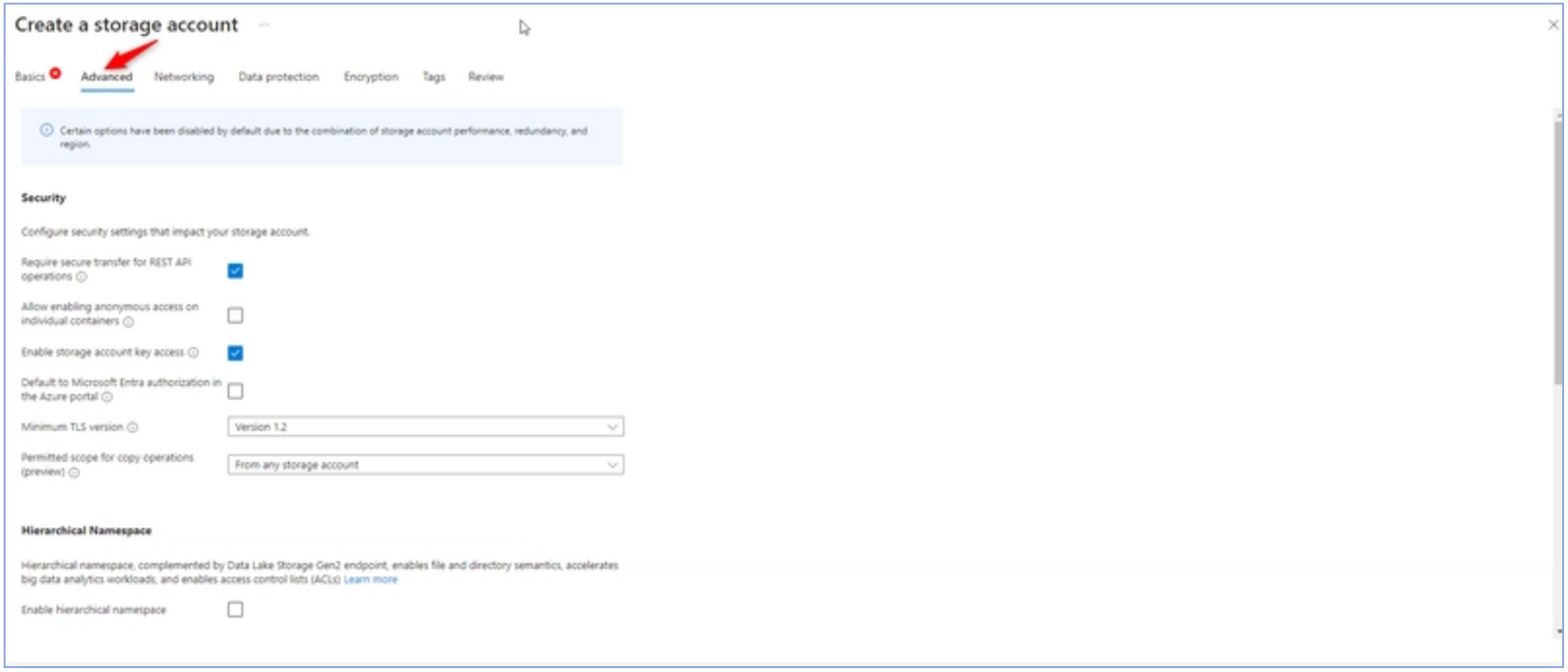
Step 4: Fill in Basic Information:

- In the "Basics" tab, provide essential details for your Storage Account:
- Subscription: Choose your Azure subscription.
- Resource group: Create a new one or select an existing one.
- Storage account name: Choose a unique name for your Storage Account.
- Region: Select the Azure region where you want to deploy your Storage Account.
- Performance: Choose between Standard and Premium performance.



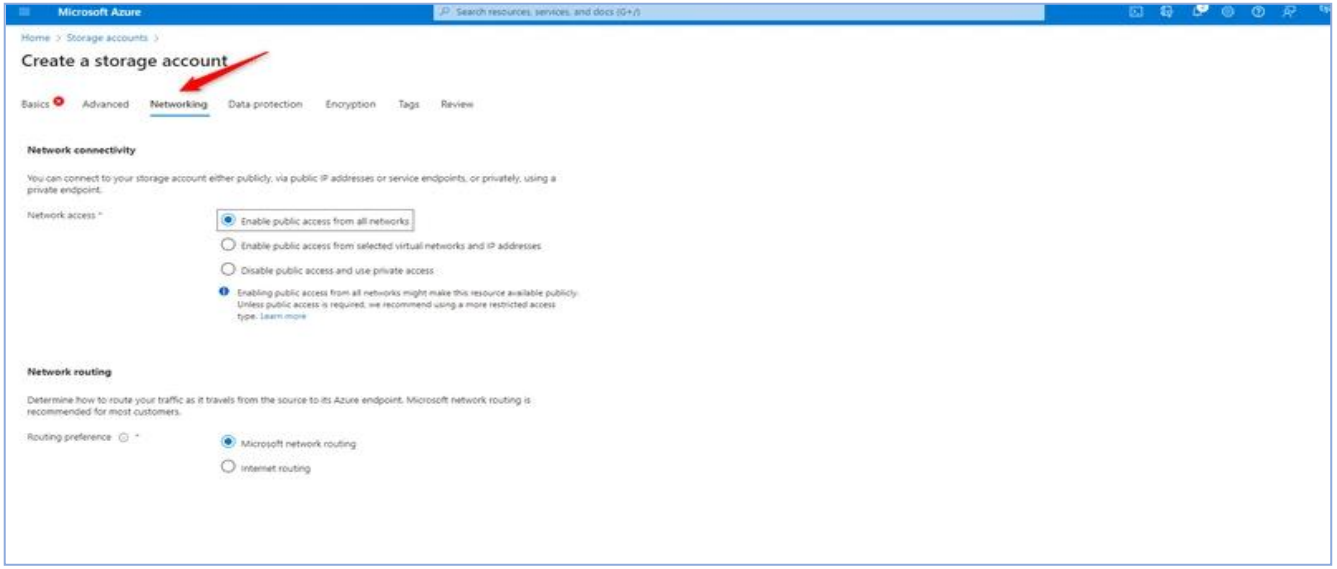
Step 5: Configure Advanced Settings:

- Navigate to the "Advanced" tab to configure additional settings:
- Account kind: Choose between StorageV2 (general-purpose v2), BlobStorage (optimized for blob storage), and others.
- Replication: Choose the replication type (e.g., Locally Redundant Storage, Zone-Redundant Storage) based on your redundancy and availability requirements.



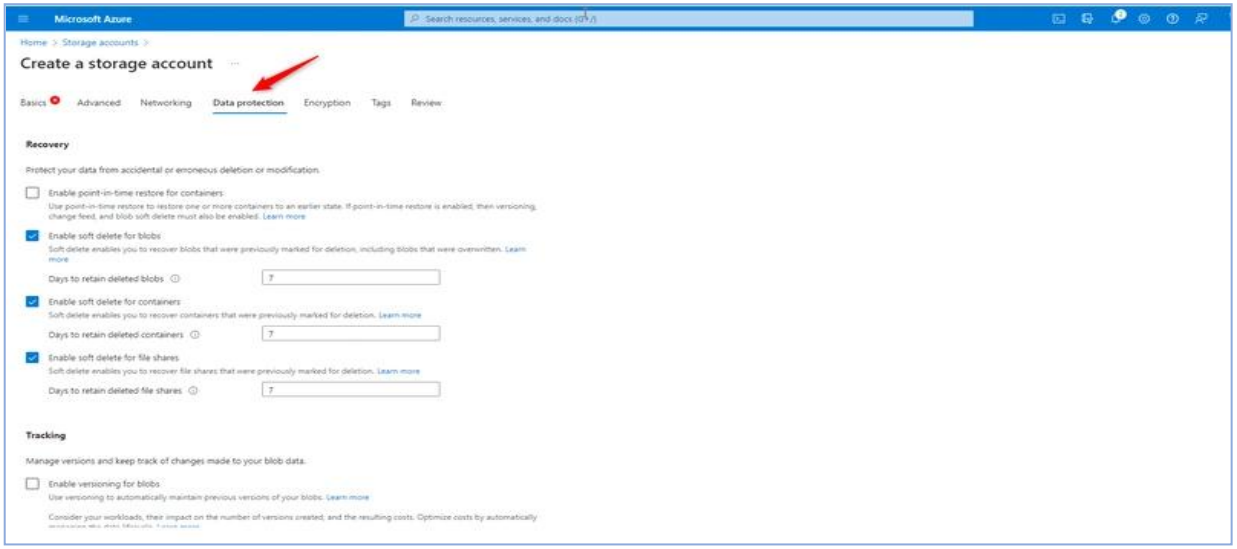
Step 6: Networking Configuration:

- In the "Networking" tab, configure network settings:
- Connectivity method: Choose how your Storage Account should be connected to networks (e.g., Public endpoint, Private endpoint).



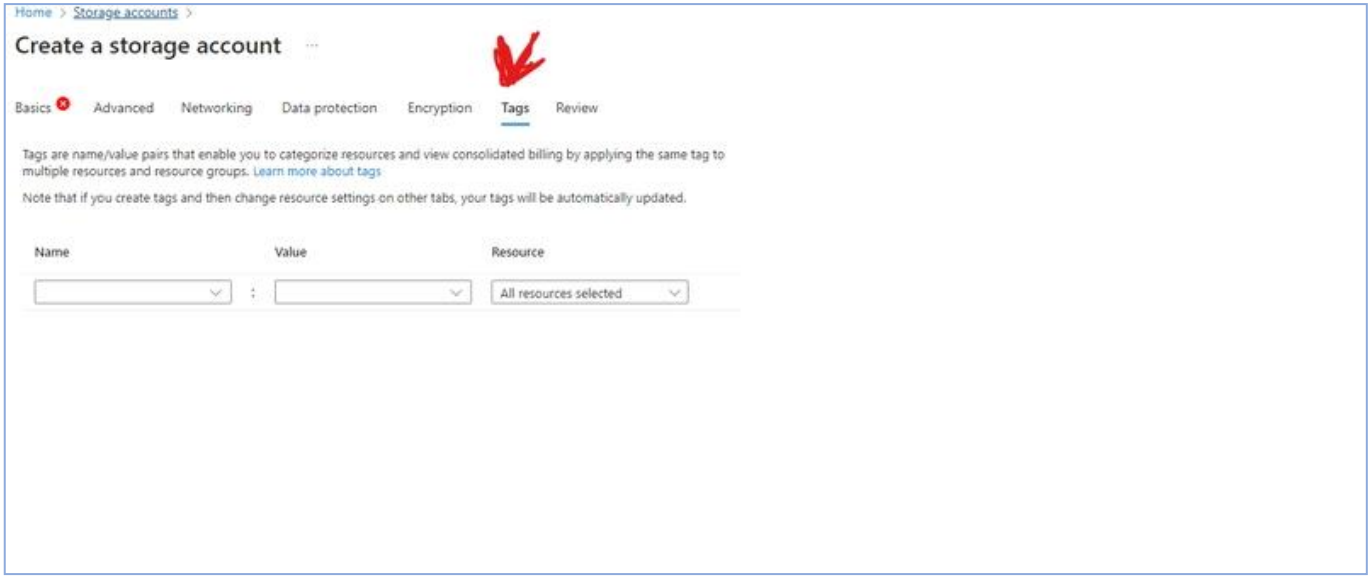
Step 7: Data Protection:

Configure data protection settings in the "Data protection" tab, such as enabling or disabling soft delete and setting retention policies.



Step 8: Tags (Optional):

You can add tags to your Storage Account for better organization and management. Tags are optional but can be useful for tracking resources.



Step 9: Review and Create:

Navigate to the "Review + create" tab to review your configurations. Click on "Create" to initiate the deployment process.

Step 10: Deployment Process:

- Azure will start deploying your Storage Account. This process may take a few moments.

Step 11: Access Your Storage Account:

- Once the deployment is complete, go to the "Storage accounts" section in the Azure Portal. Click on your Storage Account's name to view details.

Microsoft Azure

Search resources, services, and docs (G+/)

Home >

Storage accounts

Default Directory

+ Create | Restore | Manage view | Refresh | Export to CSV | Open query | Assign tags | Delete

Filter for any field... | Subscription equals all | Resource group equals all | Location equals all | Add filter

Showing 1 to 6 of 6 records.

No grouping

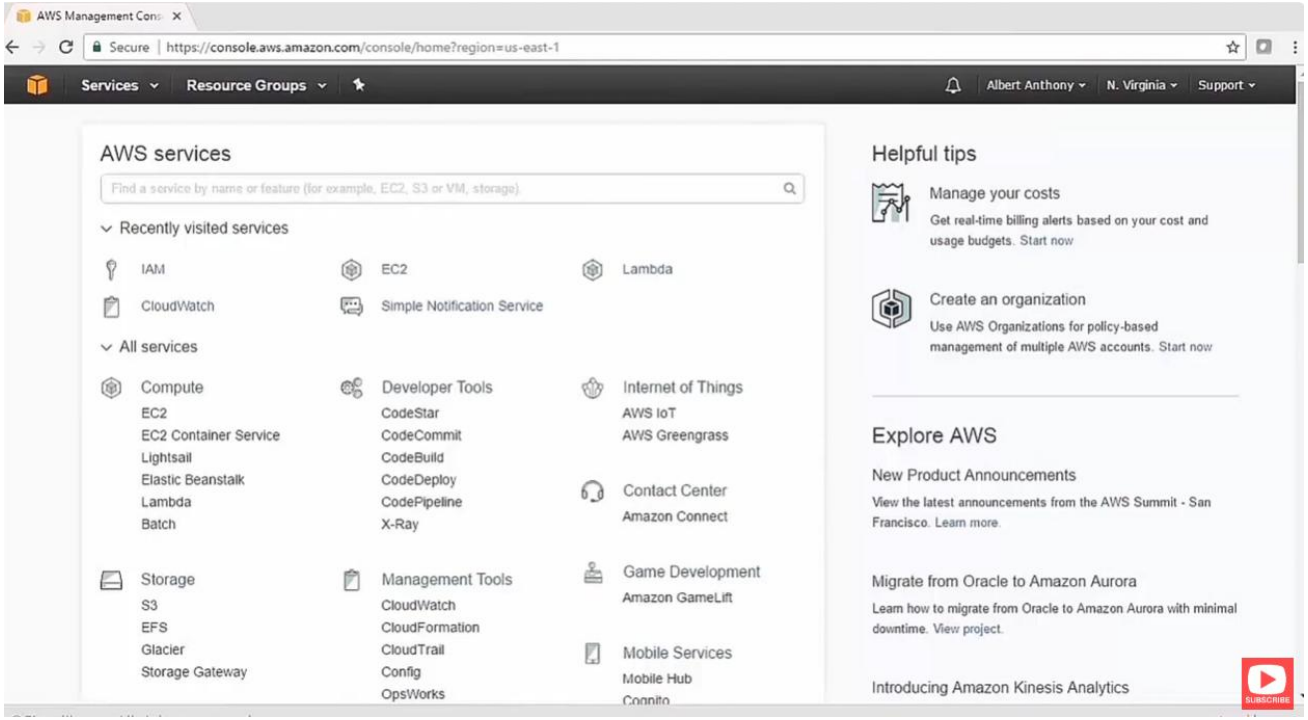
Name	Type	Kind	Resource group	Location	Subscription
companyapp	Storage account	StorageV2	publicwebsiteRG	East US	Visual Studio
companyfilesharestorage	Storage account	FileStorage	publicwebsiteRG	East US	Visual Studio
csb100320013413ae65	Storage account	StorageV2	cloud-shell-storage-westeurope	West Europe	Visual Studio
privatecompanywebsite	Storage account	StorageV2	publicwebsiteRG	East US	Visual Studio
publicwebsite40	Storage account	StorageV2	publicwebsiteRG	East US	Visual Studio
trainingstrg	Storage account	StorageV2	ulivatest	East US	Visual Studio

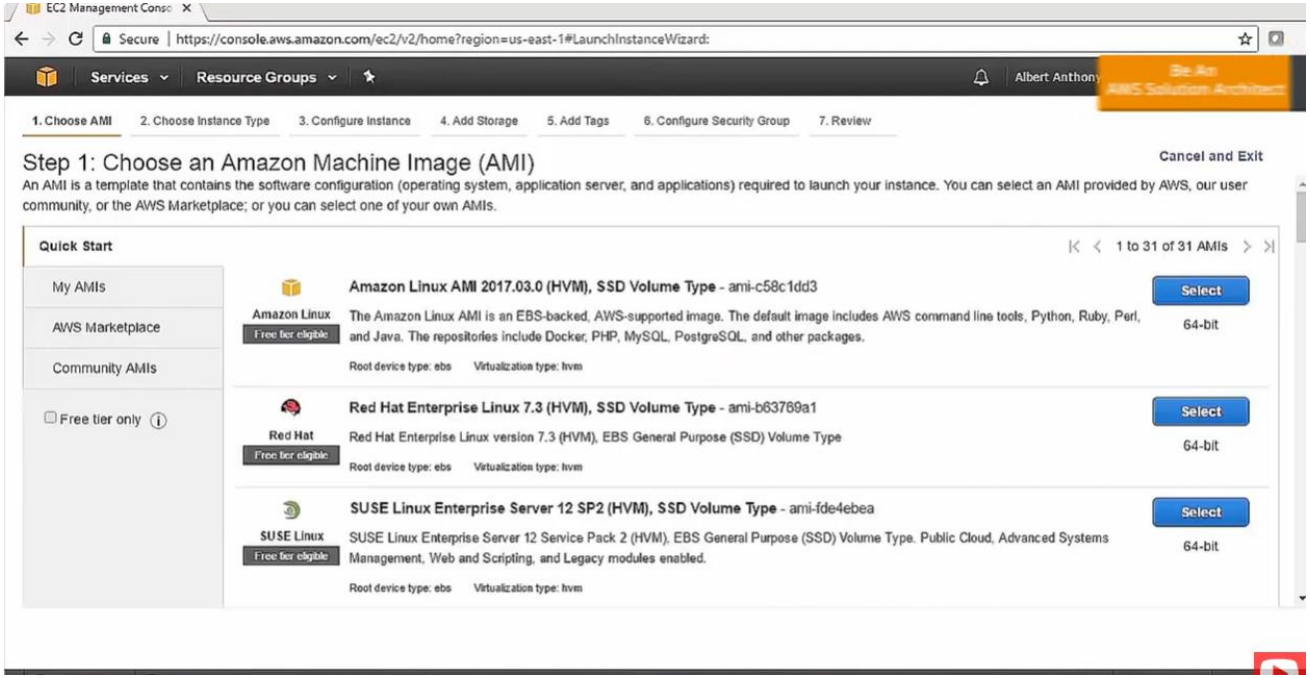
Task 13.3: Creating and Managing Compute Services

1. Create a Compute Instance (Virtual Machine)

   Steps

   I. Login to the cloud service provider console.





II. Navigate to **Compute / Virtual Machines**.

III. Click **Create Virtual Machine**.

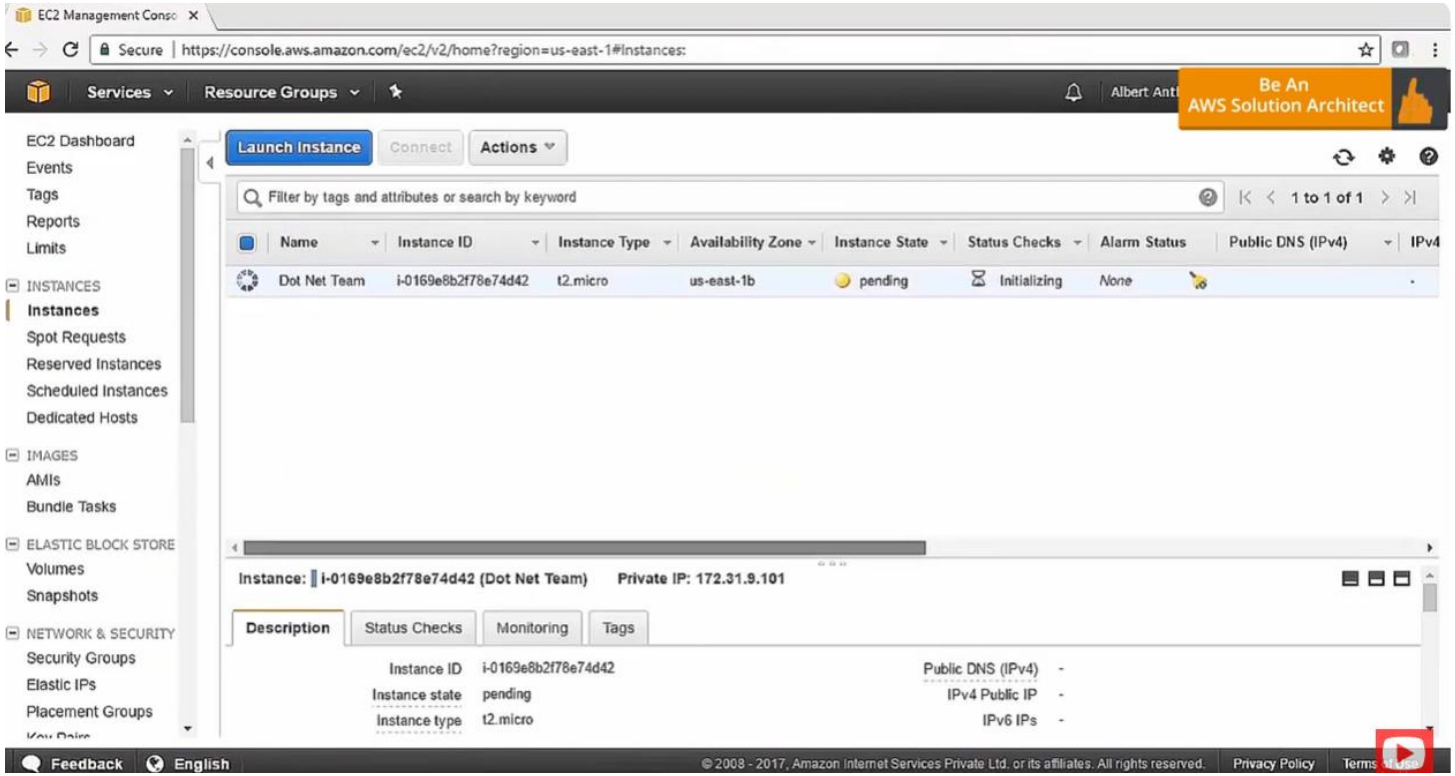
IV. Select:

   a. Operating System: Linux / Windows

   b. Instance type: Small / Free Tier

V. Click **Create**.

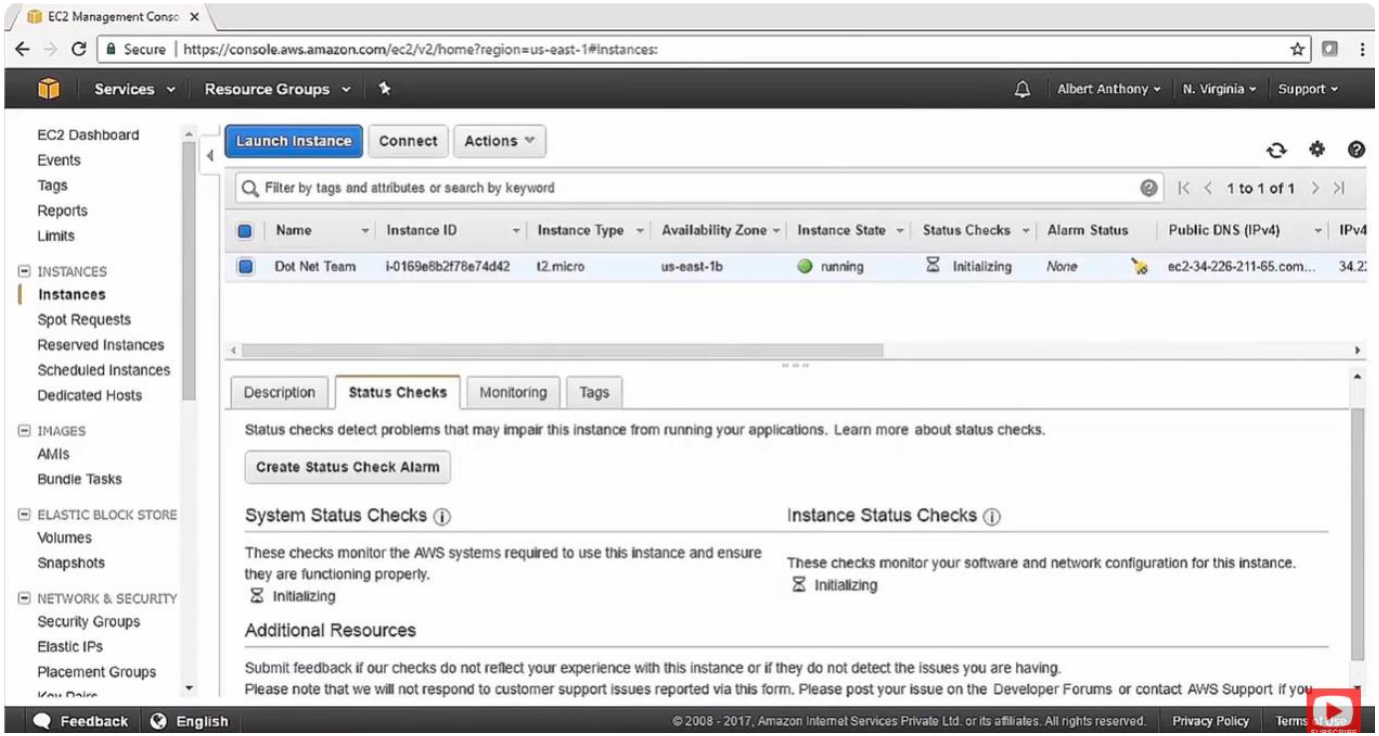
   i. ✓ The compute instance is created successfully.



2. Verify Compute Instance Running Status

Steps

1. Open the **VM dashboard**.
2. Check the status of the created VM.
3. Confirm that the status shows **Running**.
4. Note the public IP address of the VM.



3. Stop the Compute Instance to Save Cost

   Steps

   1. Select the running virtual machine.
   2. Click **Stop / Power Off**.
   3. Confirm the stop action.
   4. Observe the VM status changing to **Stopped**.

Task 13.4: Relational Database Service (RDS)

1. Create a relational database service suitable for the HealthTrack application.

   Steps

   1. Login to the cloud console.
   2. Navigate to **Database → RDS**.
   3. Click **Create Database**.
   4. Select:
      - Database Engine: **MySQL**
      - Template: **Free Tier**
   5. Enter database name (e.g., healthtrack_db).
   6. Click **Create Database**.

2. Verify that the RDS instance is active and ready to store data.

   Steps

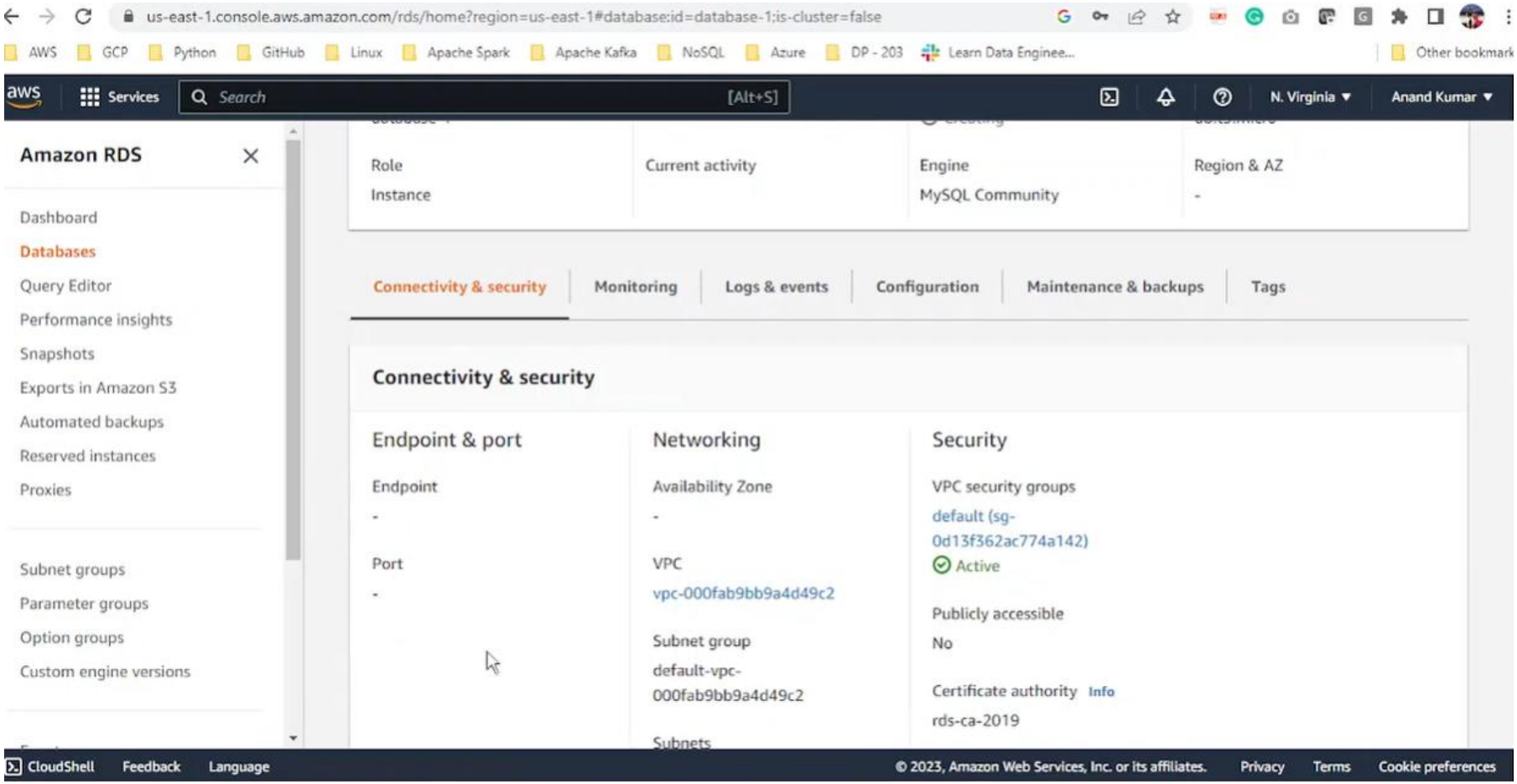
   1. Open the **RDS Dashboard**.
   2. Select the created database instance.
   3. Check the database **status**.
   4. Note the **endpoint/host name**.

      ✓ Status should show **Available**, indicating the database is ready.

3. Check Backup and Maintenance Settings

   Steps

   1. Open the database instance settings.
   2. Navigate to **Backup & Maintenance** section.
   3. Verify:
      - Automated backup is **Enabled**
      - Backup retention period is set
   4. Save settings if required.



Steps to Create an RDS Instance

https://www.knowledgehut.com/tutorials/cloud-computing/aws/creation-of-rds-instance

Following are the steps to create an RDS Instance:

1. Sign into **AWS Management Console**.
2. Open the **RDS console**.
3. In the upper-right corner, choose the region where you wish to create your instance.

4. In the navigation pane, click on '**Databases**'.
5. Click on '**Create database**'.
6. Make sure **'Standard create'** is chosen, then click on MySQL (or the database which you wish to create an RDS database instance).

Below is the snip that shows this operation.

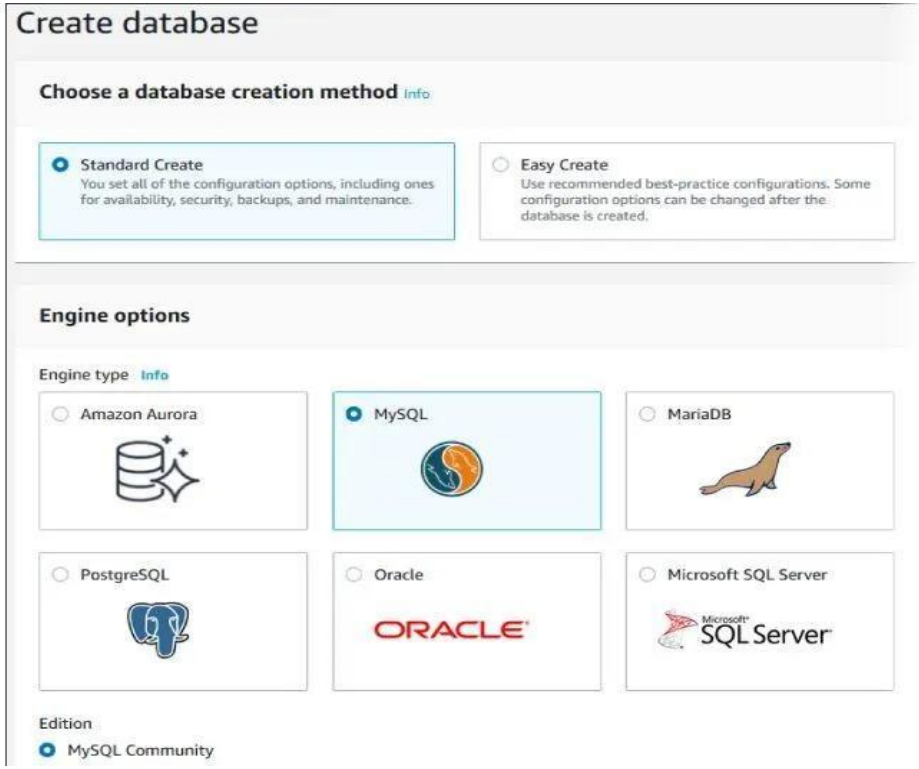


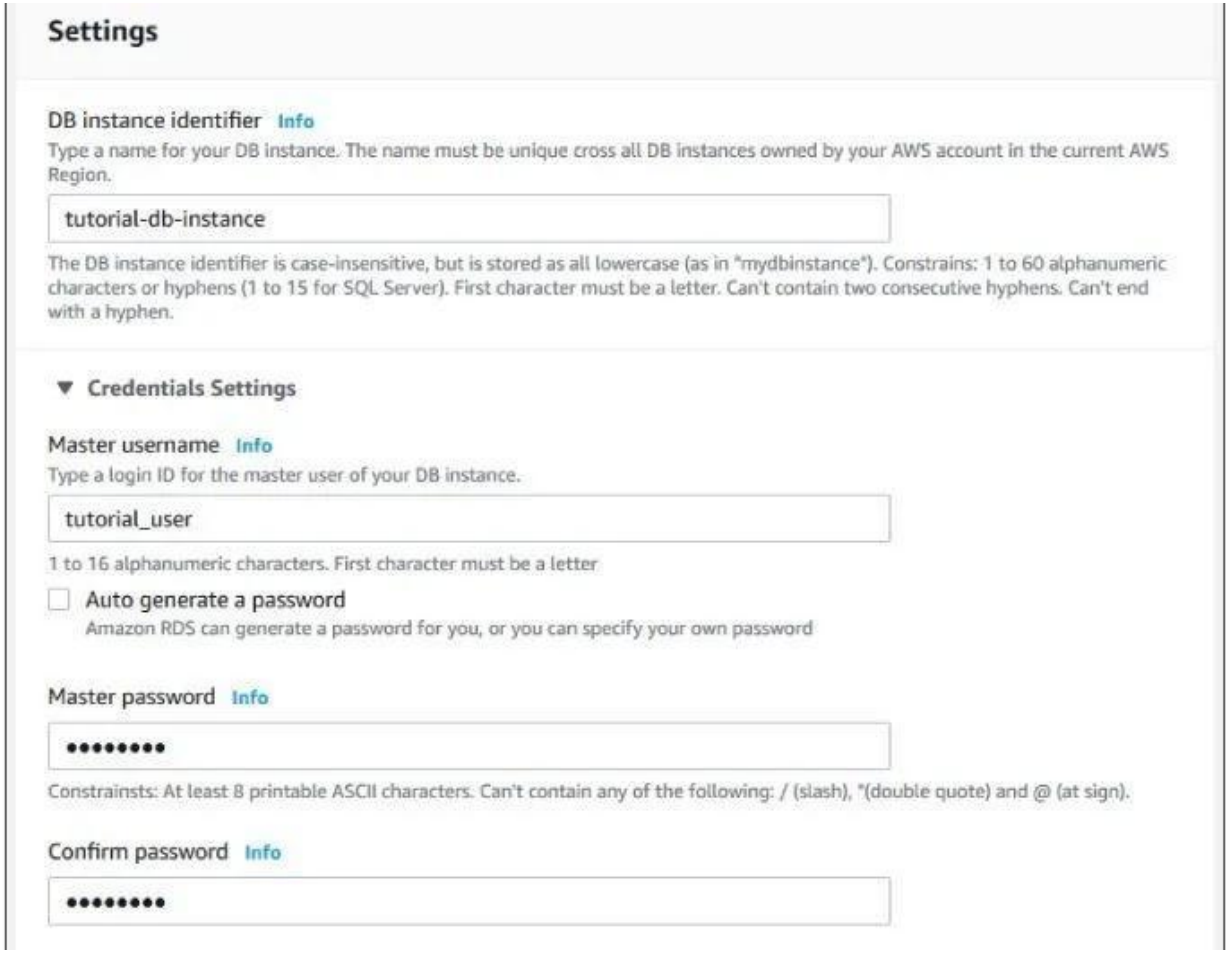
Image credit: aws.amazon.com

7. In the '**Templates**' tab, click on the '**Dev/Test**' option.
8. In the '**Setting**' tab, set the following values:

a. DB instance identifier
b. Master username
c. Auto Generate a password

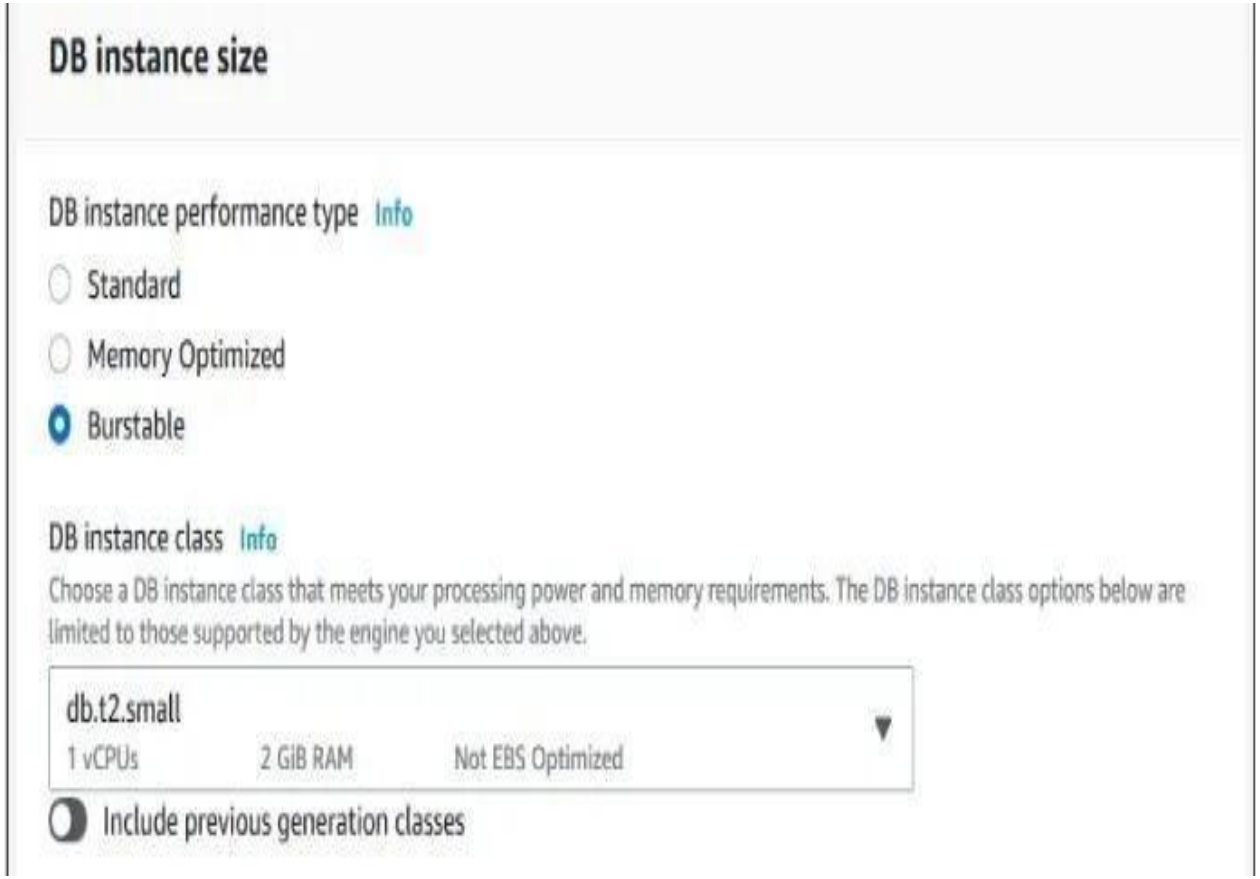
d. Master password

e. Confirm password



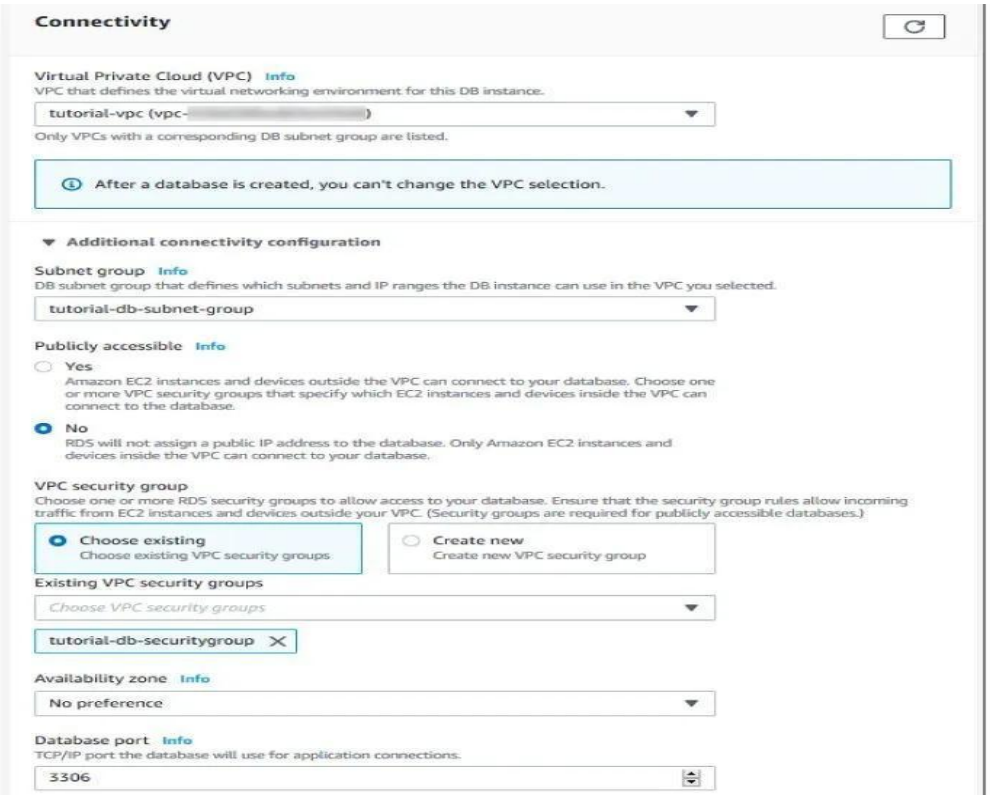
9. In the 'DB instance size' option, give a value for the following variables:

- DB instance performance types
- DB instance class

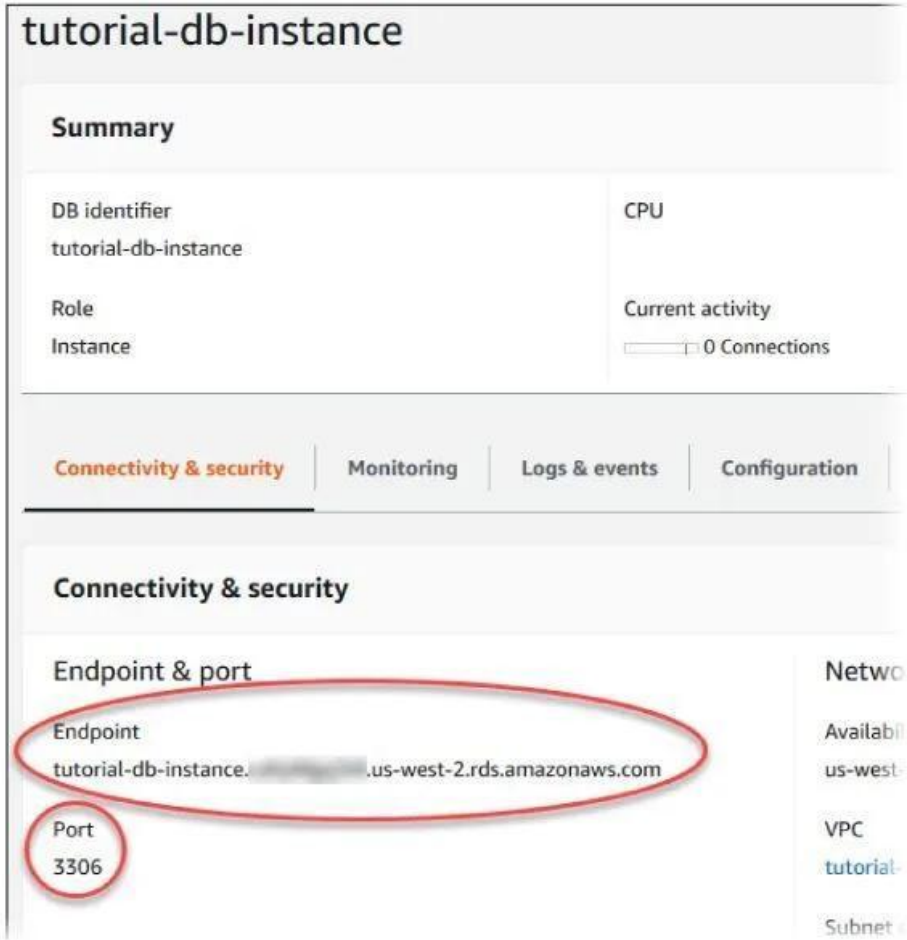


10. In the '**Storage'** and '**Availability & durability'** section, leave the default values as is.
11. In the '**Connectivity'** section, click on the 'Additional connectivity configuration' and set the below values in it:
    a. Virtual Private Cloud (VPC)
    b. Subnet group
    c. Publicly accessible- No
    d. VPC security groups
    e. Availability zone- No preference
    f. Database port- 3306

The same is displayed in the below screenshot:



12. Click on the '**Additional configuration'** tab, and provide a name for the **'Initial database name'** variable. The default settings for other options need to be kept the same.
13. Now click on '**Create database'**.
14. It takes a few minutes for the instance to get created. It can be seen in the 'Databases' list as 'Creating'.
15. Once it is created, it shows as 'Available'.
16. The '**Endpoint'** and '**Port'** of the database instance can be viewed in the 'Connectivity & security' section.



Note: Make sure that your database instance is secure, by verifying that sources outside of the VPC can't connect to the RDS database instance.

Task 13.5: Pay-As-You-Use Cloud Model

EduFlex is an online learning platform:

- **Normal days:** low traffic (≈ 200 users)
- **Exam season:** sudden spike (≈ 10,000 users)
- **Constraint:** limited capital, no heavy upfront investment

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1. Why Pay-As-You-Use is Suitable for EduFlex

Answer

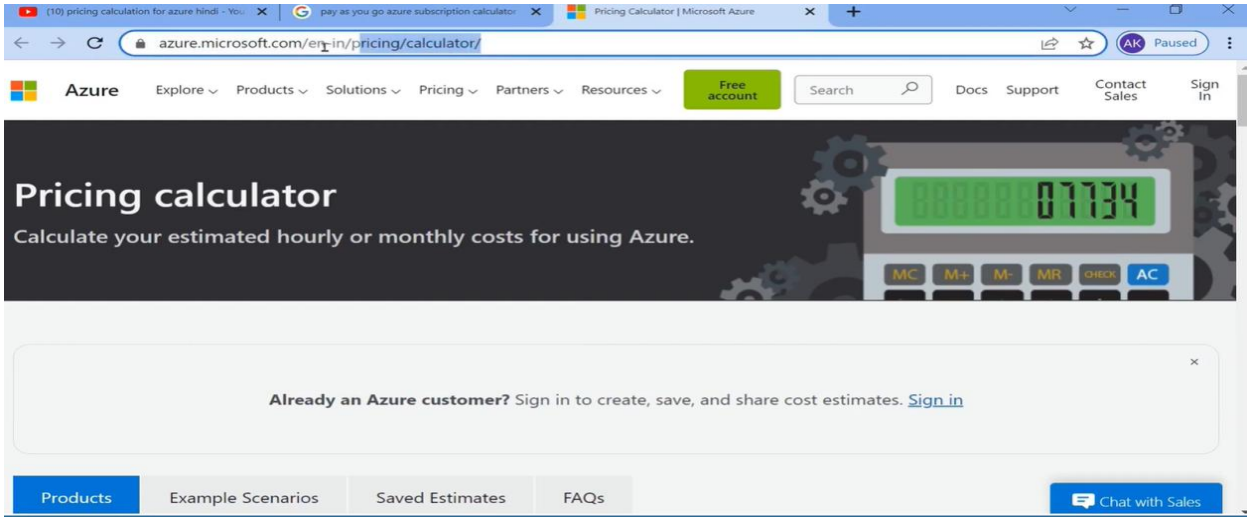
The **Pay-As-You-Use (PAYG)** model charges EduFlex **only for the resources actually used** and **only for the duration used**. This is ideal because:

- Traffic is **unpredictable**
- No need to buy servers in advance
- Costs automatically scale up/down with demand
- Perfect for startups with limited capital

Key Point: No usage → No cost.

Expected Screenshot

- Cloud **pricing page** or **pricing calculator** showing usage-based billing



2. Cloud Services EduFlex Should Use (Instead of On-Premise)

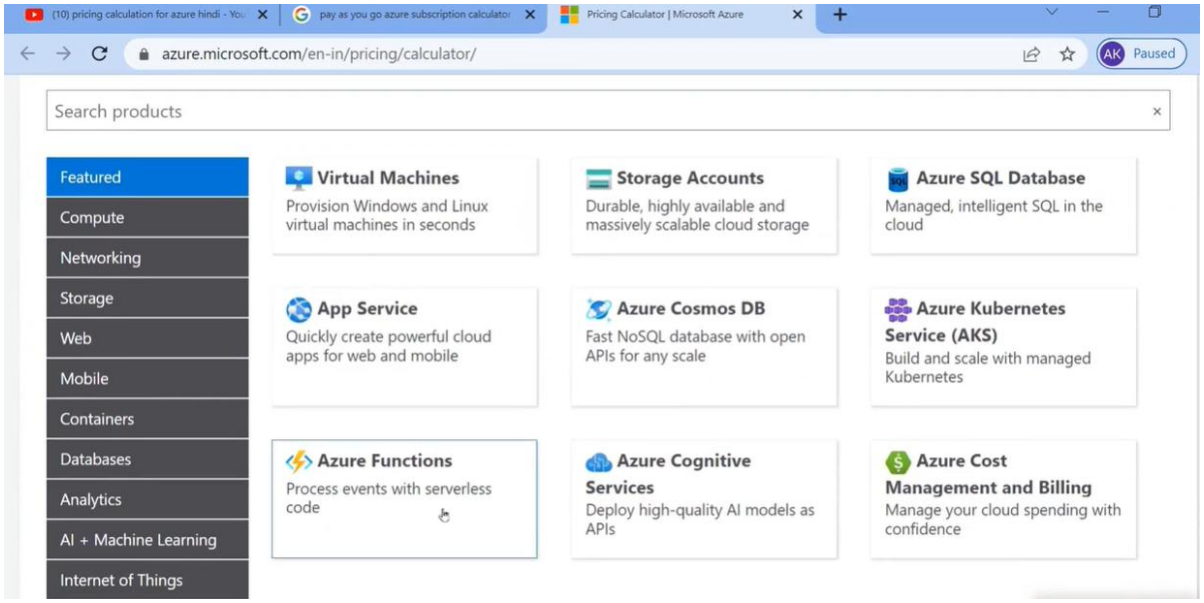
Answer

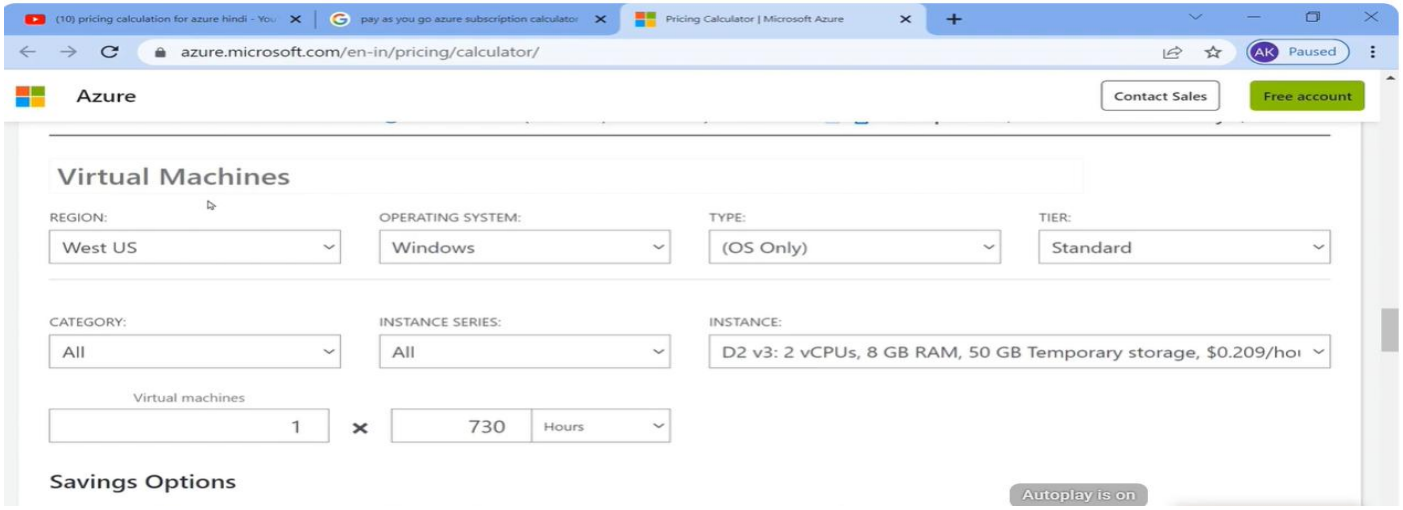
EduFlex should replace on-premise servers with the following **cloud services**:

Requirement	Cloud Service
Run web application	Compute (Virtual Machines / App Service)
Store videos, PDFs	Cloud File/Object Storage
User & course data	Managed Database (RDS)
Handle user requests	Load Balancer
Scale automatically	Auto-Scaling
Pay only for usage	Pay-As-You-Use billing

Expected Screenshot

- Cloud console home showing **Compute, Storage, Database** services





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3. Cost Benefit Estimation (Illustrative)

(Numbers are indicative for learning purposes; students should use a pricing calculator.)

Normal Days (200 users)

- 1 small VM (2 vCPU, 4 GB RAM)
- Low storage & bandwidth

Estimated Cost: Low (only minimal resources active)

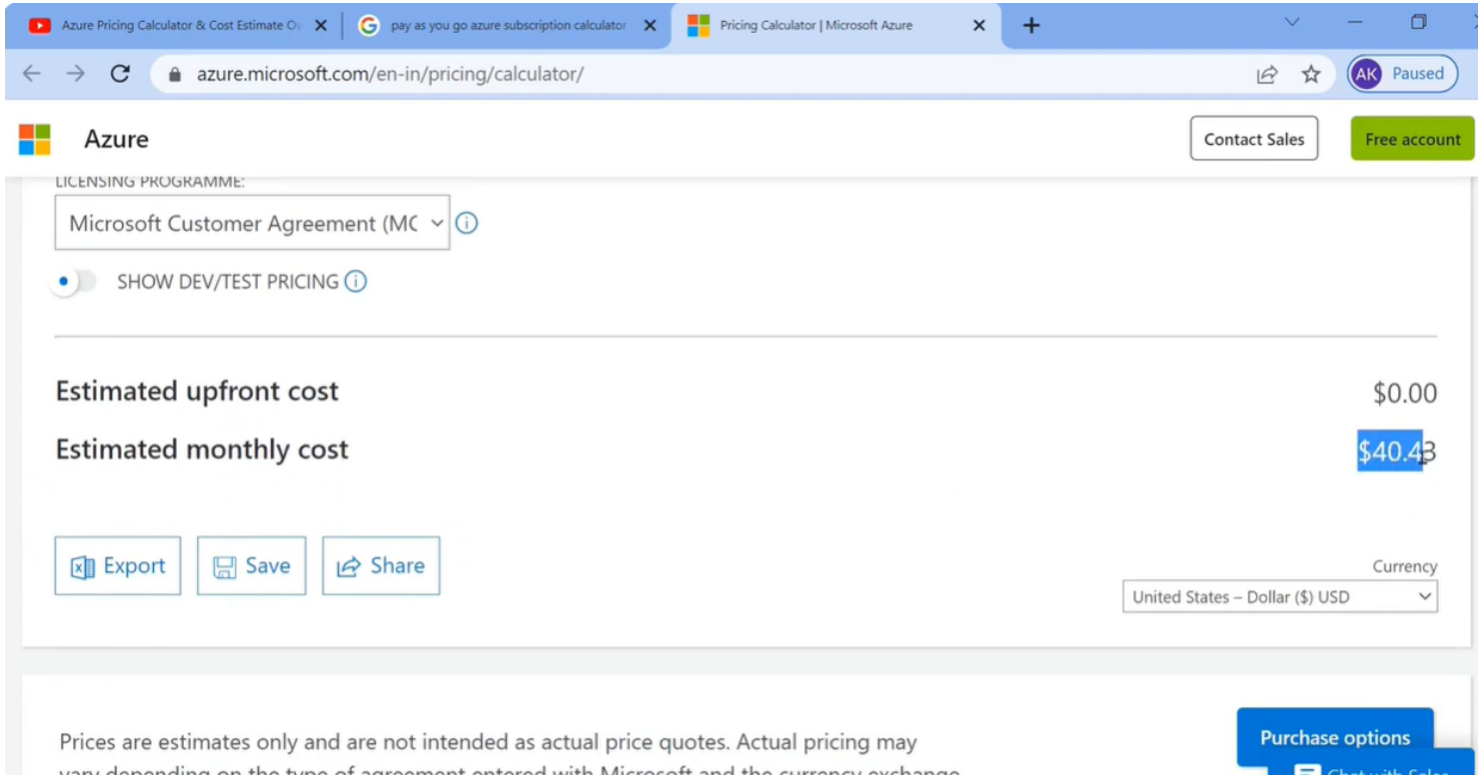
Peak Days (10,000 users)

- Auto-scaling adds multiple VMs
- Higher bandwidth and storage access

Estimated Cost: Higher **only during exam days**

Period	Infrastructure	Cost
Normal days	Minimal	Low
Exam season	Scaled up	High (temporary)
After exams	Scaled down	Low

Benefit: No payment for unused capacity.

Expected Screenshot

- Pricing calculator showing:
  - Normal usage estimate
  - Increased usage estimate (peak)